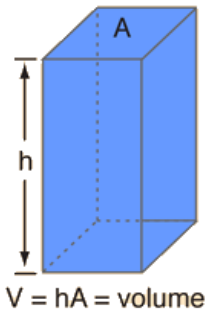
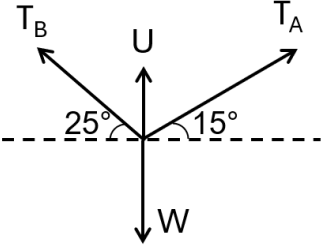


Qn	Suggested Solutions
7(a)	The weight of the fluid column $W = mg$ $= \rho Vg$ $= \rho(Ah)g$ where A is the cross-sectional area of the volume of fluid above the point  fluid column, V is the
	The hydrostatic pressure, $p = \frac{F}{A}$ $= \frac{\rho Ahg}{A}$ $= \rho gh$
(b)(i)	As pressure increases with depth and $F = PA$,
	the upward force acting on the bottom surface of the sphere is greater than the downward force acting on the top surface.
	This leads to a resultant upward force exerted by the seawater on the sphere (upthrust).
(ii)	$V_{\text{displaced}} = V_{\text{chest}}$ $= 2.0 \times 1.50 \times 0.40$ $= 1.2 \text{ m}^3$
	$U = \rho Vg$ $= (1030)(1.2)(9.81)$ $= 12125$ $\approx 1.2 \times 10^4 \text{ N}$
(iii)	$W_{\text{chest}} = \rho_{\text{chest}} Vg$ $= (1800)(1.2)(9.81)$ $= 21190$ $\approx 2.1 \times 10^4 \text{ N}$

(iv)	
	$\Sigma F_x = 0$ $T_A \cos 15^\circ = T_B \cos 25^\circ$ $T_A = 0.938 T_B \dots\dots\dots (1)$
	$\Sigma F_y = 0$ $T_A \sin 15^\circ + T_B \sin 25^\circ + U = W$ $T_A \sin 15^\circ + T_B \sin 25^\circ + 1.21 \times 10^4 = 2.12 \times 10^4 \dots\dots\dots (2)$
	$T_A = 1.28 \times 10^4 \text{ N}$
	$T_B = 1.36 \times 10^4 \text{ N}$
(c)(i)	$F_d = kv^2$ $= (3.5 \times 10^3)(4.5)^2$ $= 70875$ $\approx 7.09 \times 10^4 \text{ N}$
	Constant velocity means net force = 0 N $F_{\text{thrust}} = F_d$ $= 7.09 \times 10^4 \text{ N}$
(ii)	$P = \frac{E}{t}$ $= \frac{Fs}{t}$ $= Fv$
	$P = F_{\text{thrust}} v$ $= (70875)(4.5)$ $= 3.19 \times 10^5 \text{ W}$
	$\eta = \frac{\text{useful output power}}{\text{input power}} \times 100\%$ $= \frac{3.19 \times 10^5}{0.50 \times 10^6} \times 100\%$ $= 63.8\%$ $\approx 64\%$

(d)(i)	$F_{\text{net}} = \text{force removed}$ $= (200)(9.81)$ $= 1962 \text{ N}$
	total mass = mass of submarine + mass of box $= (3600 - 200) + (1800)(1.2)$ $= 5560 \text{ kg}$
	$\Sigma F = ma$ $1962 = (5560)a$ $a = 0.353 \text{ m s}^{-2}$
(ii)	Due to the acceleration, speed increases which causes drag force to increase until
	drag force is equal to the difference between upthrust and the reduced weight. OR Net upward force and acceleration becomes zero. (The submarine then moves at a constant "terminal velocity".)

Anglo-Chinese Junior College

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